

PhishGPT: Automating HTML Phishing Emails with ChatGPT

Author: Marcelo Ribero Cerro

Supervisor: dr. H.K. Schraffenberger

Second Reader: dr. M. Santos Wessel

INTRODUCTION & BACKGROUND

INTRODUCTION & BACKGROUND

EMAILS: PHISHING



Evan Ston
To

Thu 12/10/2020 10:36 AM

 You forwarded this message on 12/10/2020 10:39 A.M.

Hello,

I am a staff in the college, a professor of Medicine shared me a link for students who might be interested in a PAID UNICEF PART TIME POSITION job to make up to \$500 weekly,

Email for More info - freemanjohnny6s66@outlook.com

NOTE: This is strictly Work From Home Position

INTRODUCTION & BACKGROUND

EMAILS



INTRODUCTION & BACKGROUND

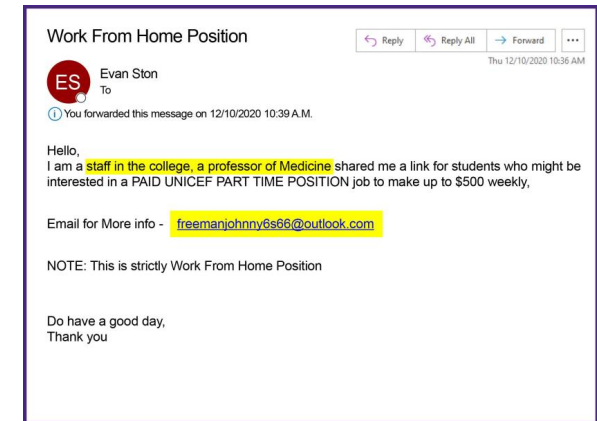
EMAILS: COMPARISON



Visual aspect has gained a lot of importance



Created expectations on receivers



INTRODUCTION & BACKGROUND
LARGE LANGUAGE MODELS



Code generation



Text generation



ChatGPT

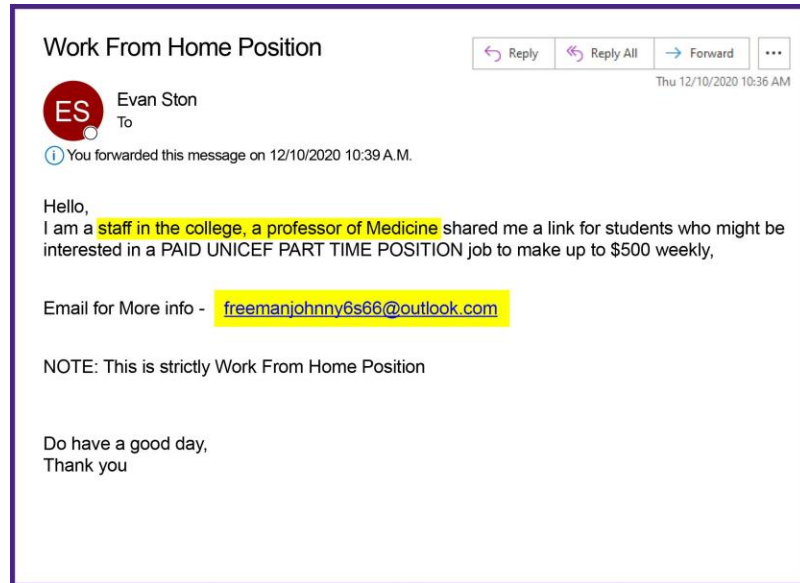
 Gemini

 Claude

 deepseek

INTRODUCTION & BACKGROUND

PREVIOUS RESEARCH



Text generation



Text generation & Code generation

METHODOLOGY

RESEARCH DESIGN: QUESTIONS

RESEARCH QUESTION

- Can LLMs generate **convincing** phishing HTML emails that accurately mimic authentic professional communications?

RESEARCH DESIGN

PROGRAM DESIGN



MAKE A PROGRAM THAT
AUTOMATES THE CREATION OF
HTML EMAILS USING CHATGPT

SURVEY DESIGN



CONDUCT A USER SURVEY TO
ANALYZE QUALITY OF THE
RESULTS

RESULT ANALYSIS



ANALYZE THE RESULTS AND
DRAW CONCLUSIONS

METHODOLOGY: PROGRAM DESIGN

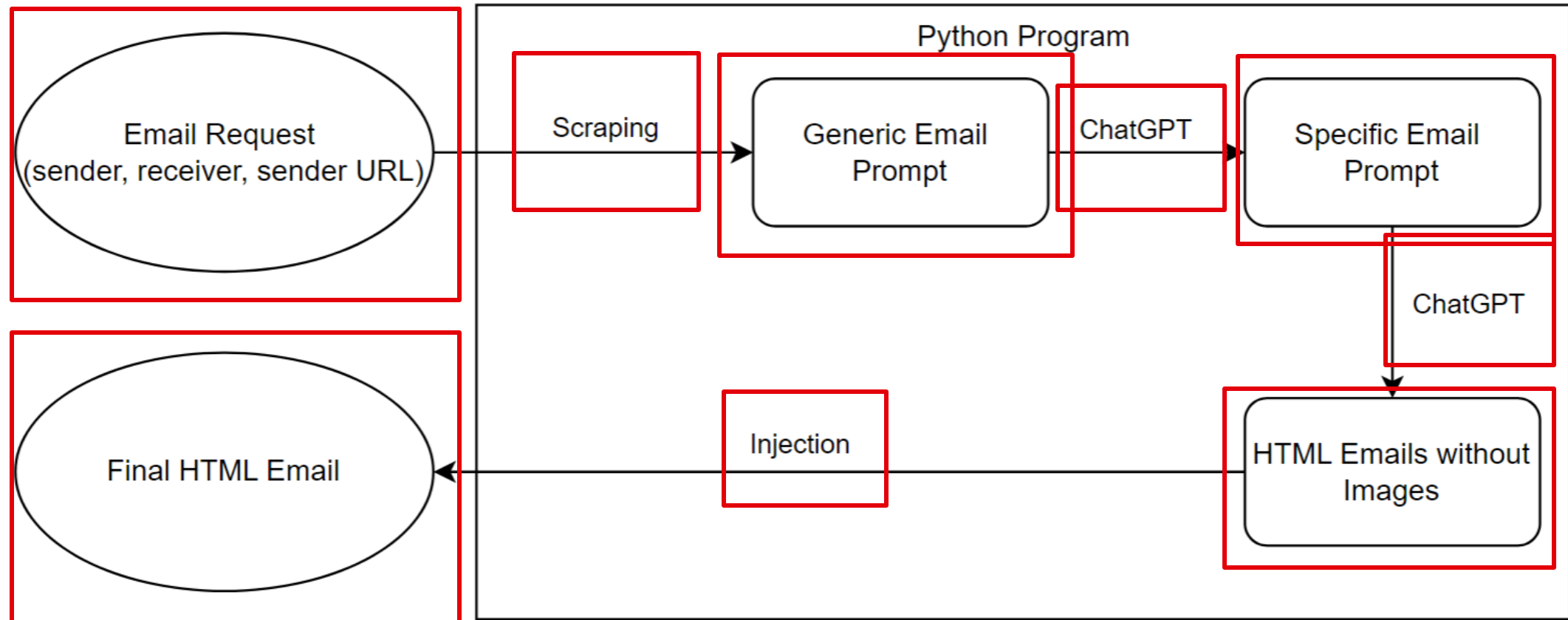
PROGRAM GOAL

Goal: For an arbitrary input (sender, receiver, topic, URL) $\rightarrow_{generate}$ HTML email

sender: entity being impersonated
receiver: target recipient
topic: subject or theme of the email
URL: URL of the sender

METHODOLOGY: PROGRAM DESIGN

PROGRAM FLOW GRAPH



METHODOLOGY: SURVEY DESIGN

EMAILS INCLUDED

- Airbnb
- Banking
- Bolt
- Booking
- Decathlon
- H&M
- NordVPN
- Ryanair
- Uber eats
- Google

Two versions

Authentic version

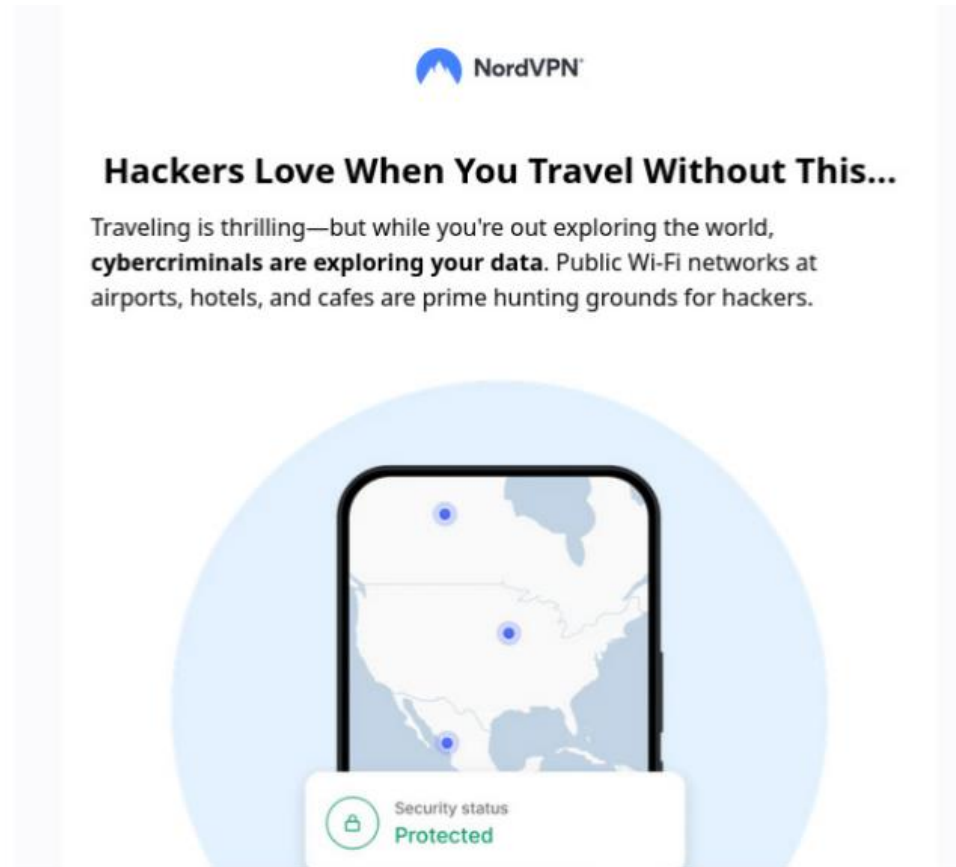
Obtained from a real inbox

Fabricated version

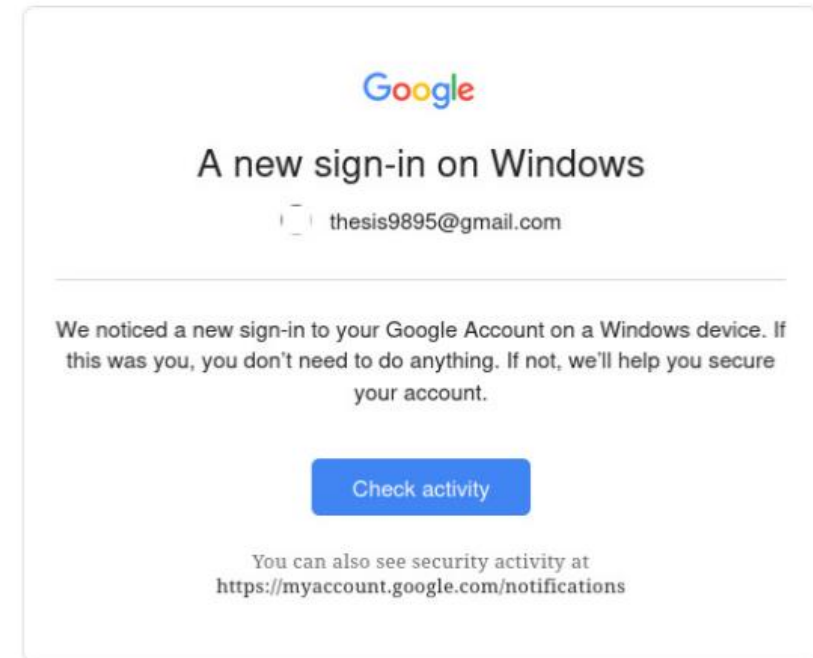
Created by our program

METHODOLOGY: SURVEY DESIGN

MOTIVATION: HIGH INFLUENCE AND LOW INFLUENCE



Example of low-influence email



You received this email to let you know about important changes to your Google Account and services.

© 2025 Google Ireland Ltd., Gordon House, Barrow Street, Dublin 4, Ireland

Example of high-influence email

METHODOLOGY: SURVEY DESIGN
NO PHISHING EMAILS

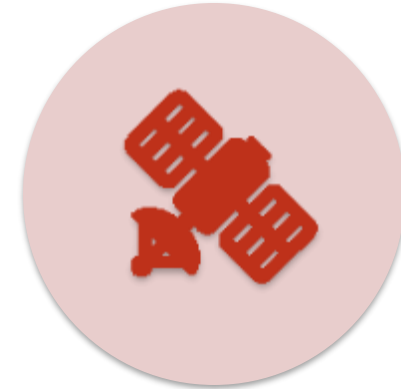
LACK OF IMPORTANT CLUES



NO TOPIC RELEVANCE



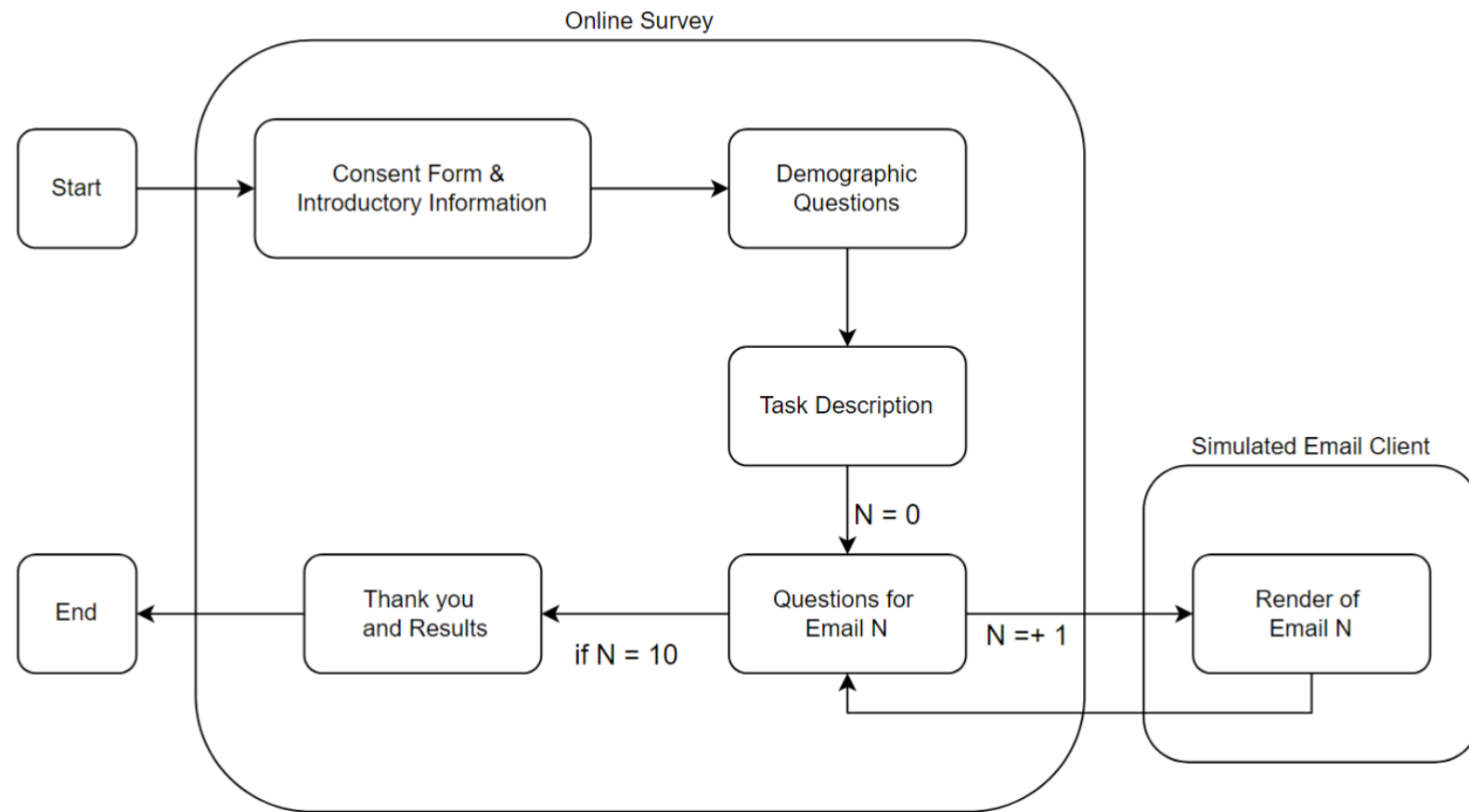
NO EMAIL ADDRESS



NO MALICIOUS PAYLOAD

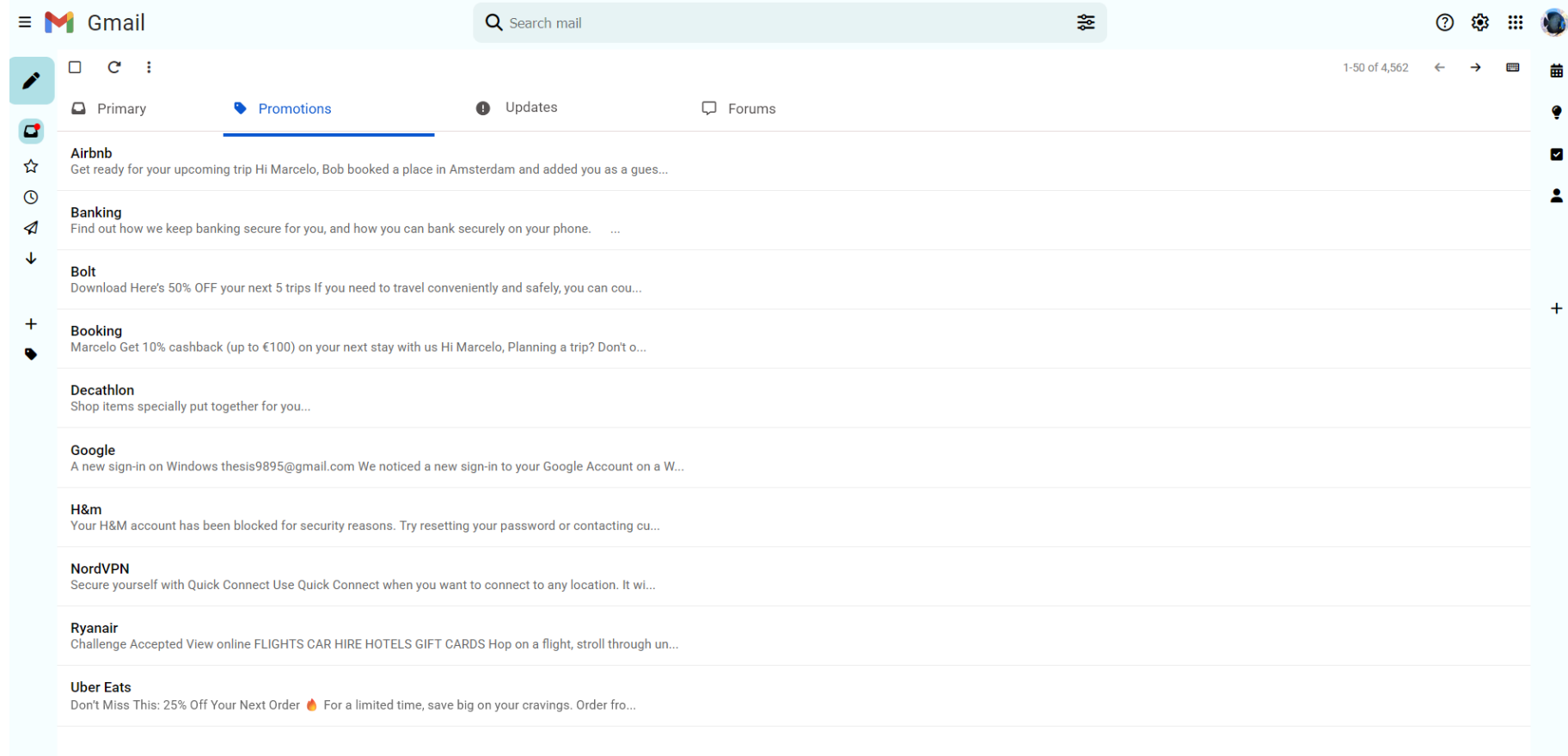
METHODOLOGY: SURVEY DESIGN

SURVEY FLOW GRAPH



METHODOLOGY: SURVEY DESIGN

SIMULATED EMAIL CLIENT



EXAMPLE 1

1. Do you think this email was generated by ChatGPT?
2. Do you think this is a phishing email?
3. How confident are you in your classifications? Both ChatGPT and Phishing
4. How motivated were you to click the link?
5. Explain your reasoning behind your classifications



Discover Our New Collection!

Dear fashion enthusiast,

We're thrilled to unveil our latest collection at H&M! Dive into a world of style with exclusive designs that redefine chic. Every piece is crafted to ensure you step out with confidence and flair. Stay ahead of the fashion curve with our new arrivals.

Featured Styles



Experience the allure of our vibrant patterns and premium fabrics, designed to make heads turn. From bold prints to subtle elegance, this collection caters to every style aspiration.

What Customers Love

"Absolutely in love with the new collection! The quality is top-notch, and the styles are simply iconic. Can't wait to add more pieces to my wardrobe!" - *Sophia, Fashion Blogger*

EXAMPLE 2

1. **Do you think this email was generated by ChatGPT?**
2. Do you think this is a phishing email?
3. How confident are you in your classifications? Both ChatGPT and Phishing
4. How motivated were you to click the link?
5. Explain your reasoning behind your classifications



Get More Out of Your Credit Card – Discover the Benefits

Dear Marcelo,

Have you ever wondered how to make the most of your credit card? IGN Netherlands Bank offers a wide range of benefits that can make your everyday purchases even more rewarding. Discover how you can take full advantage of the exclusive perks and rewards waiting for you.

Exclusive Benefits Just for You

From cashback on everyday purchases to exclusive discounts at your favorite stores, the benefits of using your IGN Netherlands Bank credit card are designed to enhance your shopping experience. Enjoy access to:

- Cashback rewards on eligible purchases
- Discounts at over 500 partner merchants
- Easy online payments with added security

Start Earning Today

The process is simple! Just use your IGN Netherlands Bank credit card on your daily purchases and watch the benefits stack up. Whether you're buying groceries, booking flights, or enjoying a night out, you'll be rewarded.

Unlock Your Benefits Now

RESULTS & DISCUSSION

RESULTS & DISCUSSION

OVERVIEW

The total amount of participants was 67

Each participant classified 10 emails

Total of emails classified 670

Classification			Count	Total %
ChatGPT	Type	Authentic	348	51.9%
		Fabricated	322	48.1%
	Accuracy	Correctly Classified	393	58.7%
		Incorrectly Classified	277	41.3%
Phishing	Type	Phishing	261	38.9%
		Not Phishing	409	61.1%
	Accuracy	Real Phishing Emails	0	—

RESULTS & DISCUSSION

OVERVIEW

Category	Count	Percentage
Total fabricated emails	322	100%
Classified as phishing	114	35.40%
Classified as authentic	156	48.45%
Classified as authentic and not phishing	111	34.47%

Classification			Count	Total %
ChatGPT	Type	Authentic	348	51.9%
		Fabricated	322	48.1%
	Accuracy	Correctly Classified	393	58.7%
		Incorrectly Classified	277	41.3%
Phishing	Type	Phishing	261	38.9%
		Not Phishing	409	61.1%
	Accuracy	Real Phishing Emails	0	—

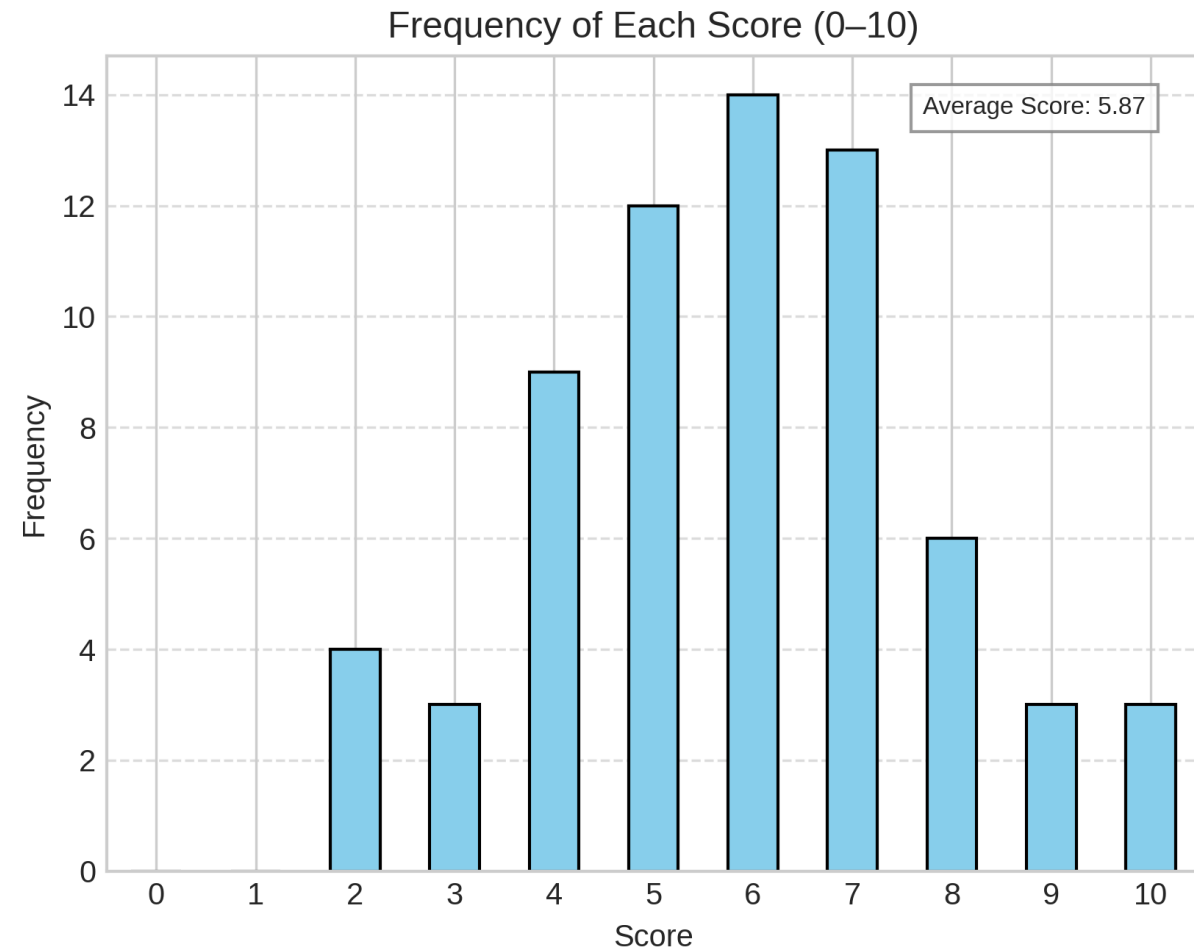
RESULTS & DISCUSSION

FREQUENCY OF SCORES

Hypothesis testing results:

- Testing showed evidence of the mean score being different from chance

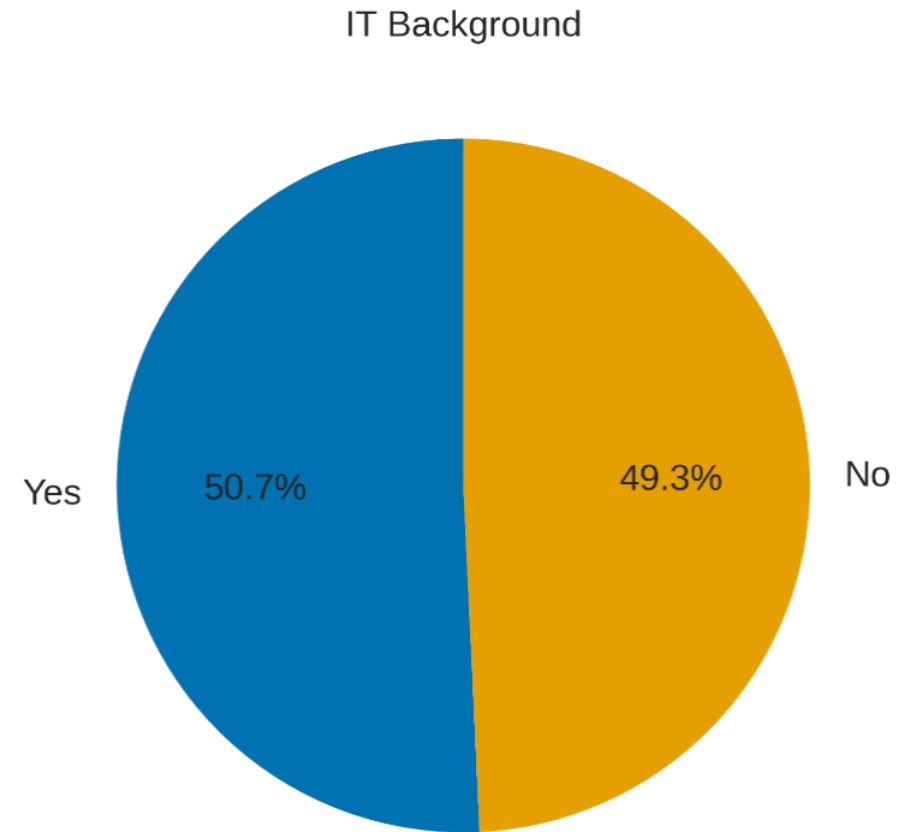
Metric	Value
Classification Score (0-10)	
Mean score	5.87
Range	8 (Min: 2, Max: 10)
Median	6
Mode	6
Standard deviation	1.95
1st Quartile (Q1)	5.00
3rd Quartile (Q3)	7.00
Confidence Rating (0-4 scale)	
Mean confidence	2.58
Standard deviation	0.66



FREQUENCY OF SCORES

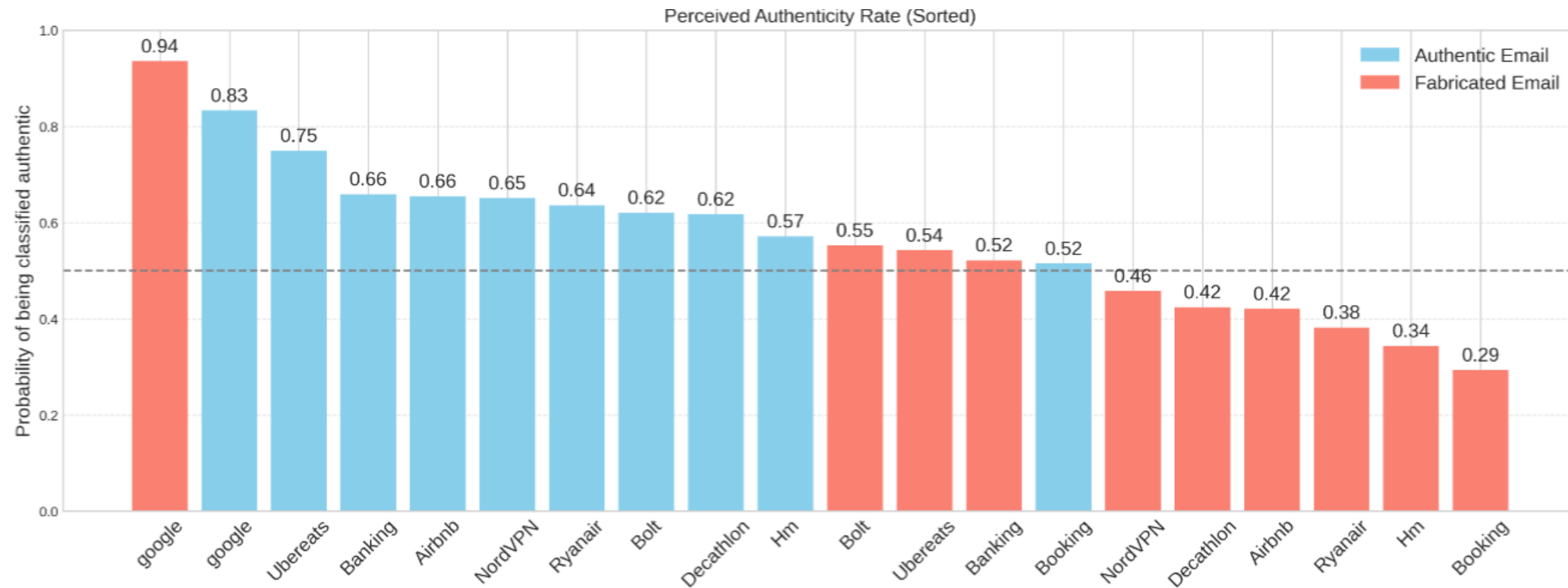
Hypothesis testing results:

- Testing showed no evidence that mean score between participants with and without CS/IT background



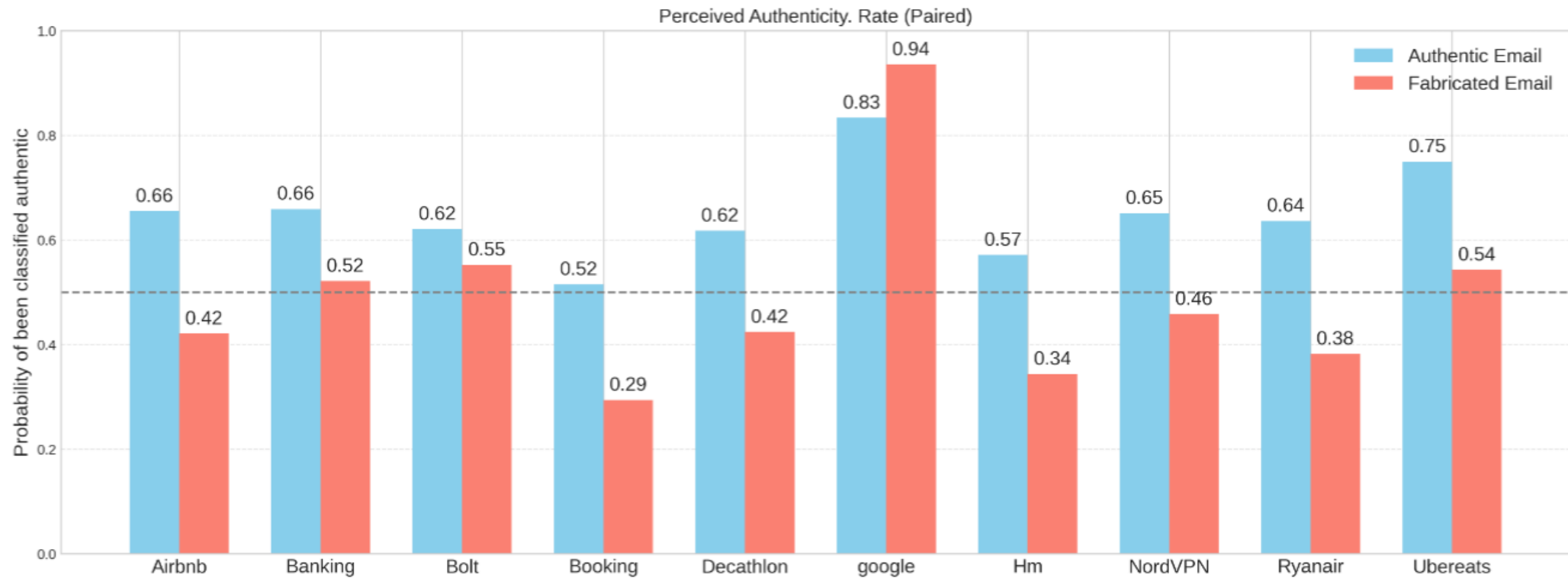
RESULTS & DISCUSSION

CHATGPT CLASSIFICATION



RESULTS & DISCUSSION

CHATGPT CLASSIFICATION

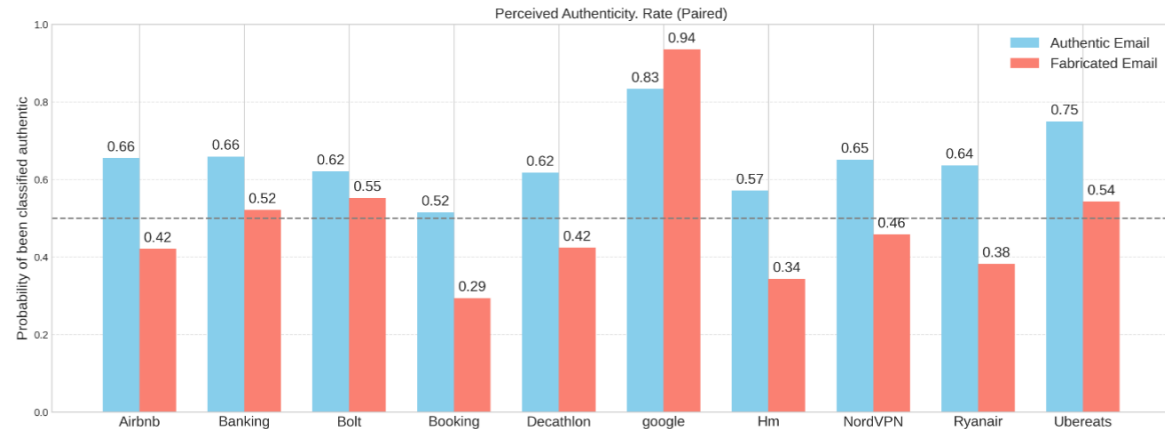
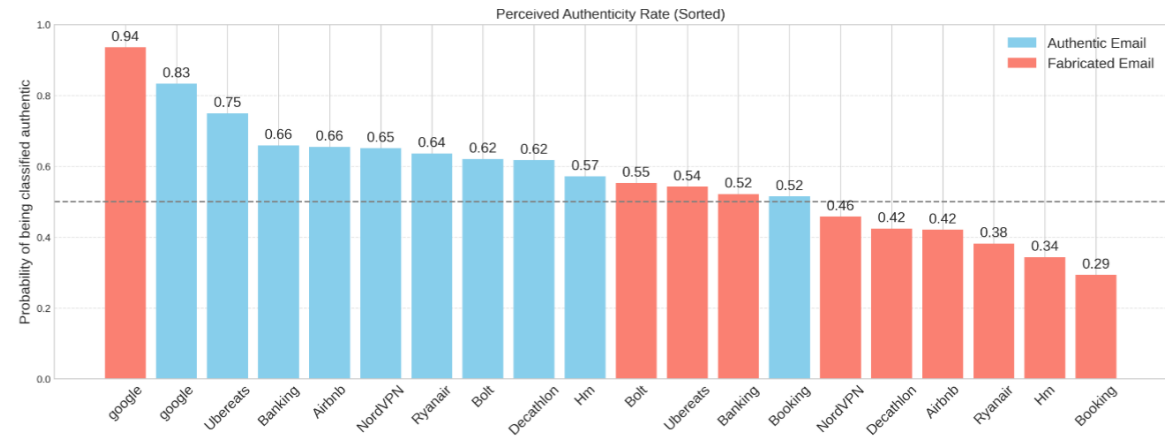


RESULTS & DISCUSSION

CHATGPT CLASSIFICATION

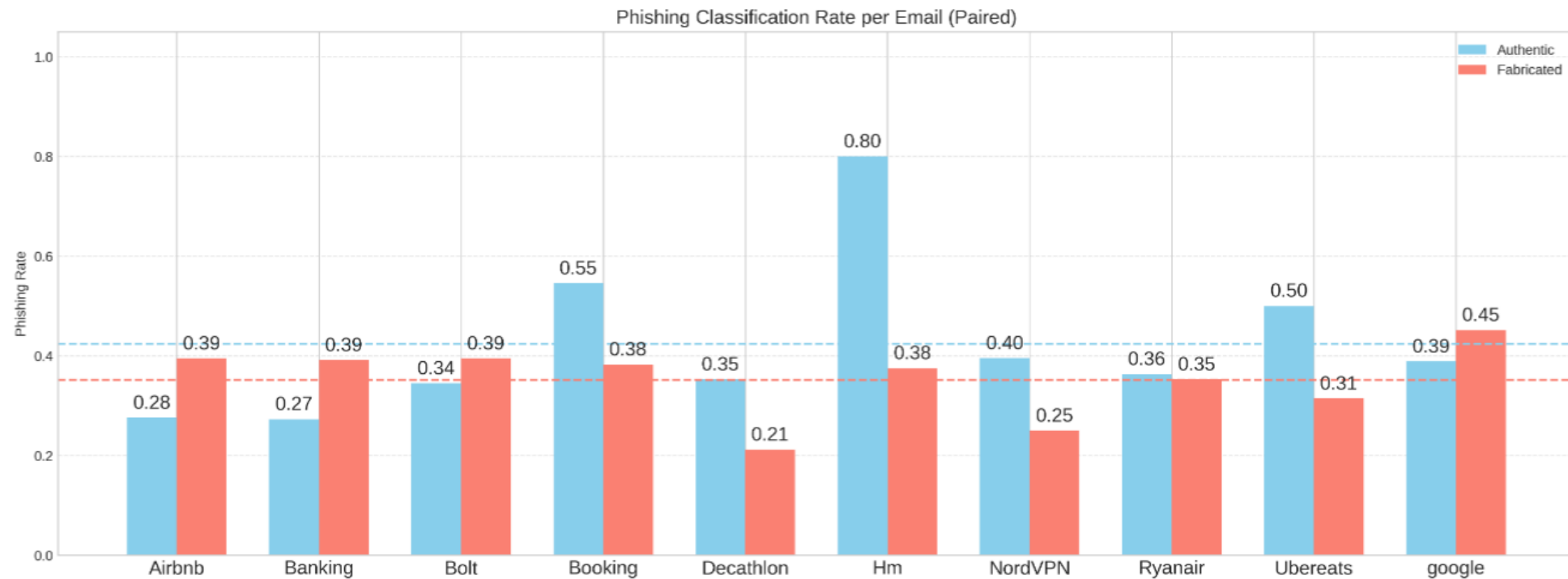
Hypothesis testing results:

- Testing showed evidence of the mean accuracy rates of fabricated emails and authentic emails being different
- Testing showed no evidence that fabricated emails' accuracy is different from chance



RESULTS & DISCUSSION

PHISHING CLASSIFICATION

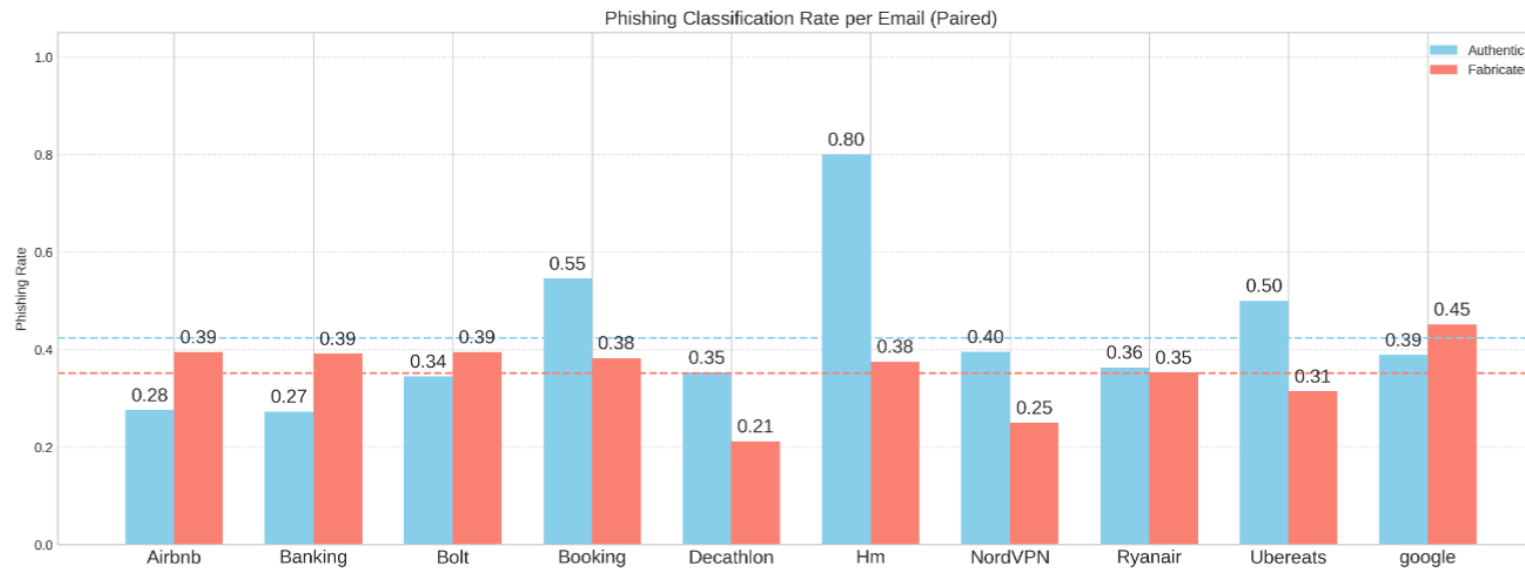


RESULTS & DISCUSSION

PHISHING CLASSIFICATION

Hypothesis testing

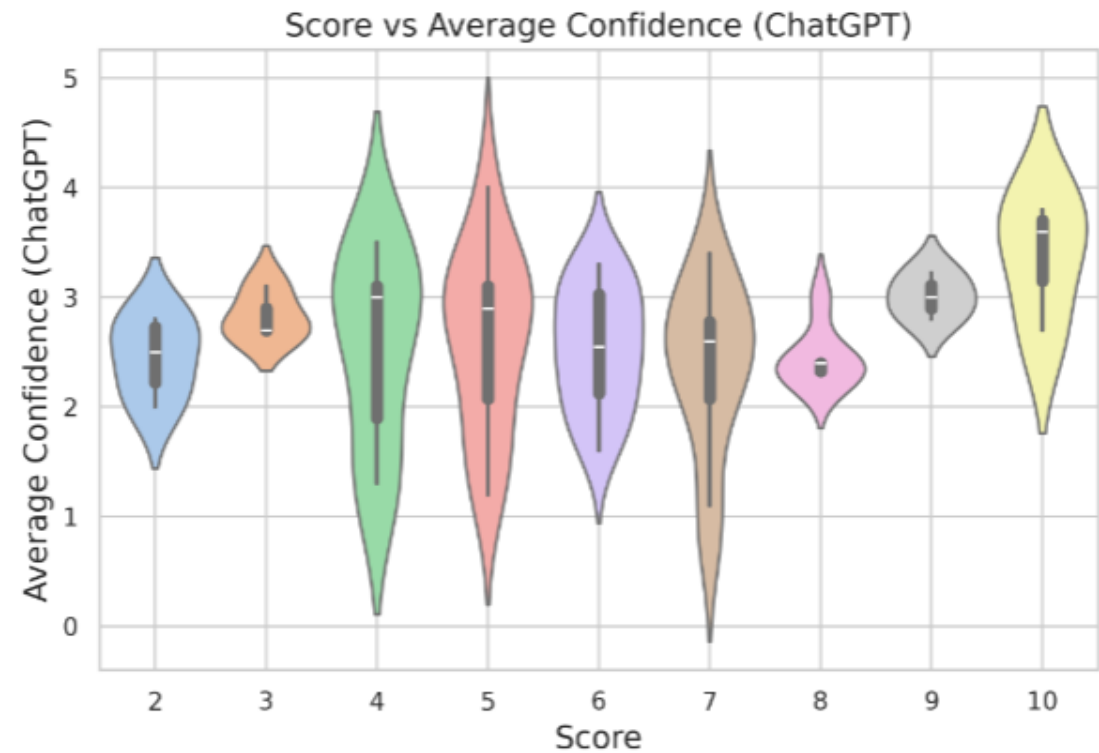
- Testing showed no evidence that phishing classification differs in fabricated and authentic emails
- Testing showed no evidence of a relation between ChatGPT and Phishing classification



SCORE & CONFIDENCE

Hypothesis testing results:

- Testing showed no evidence of a linear correlation between score and confidence

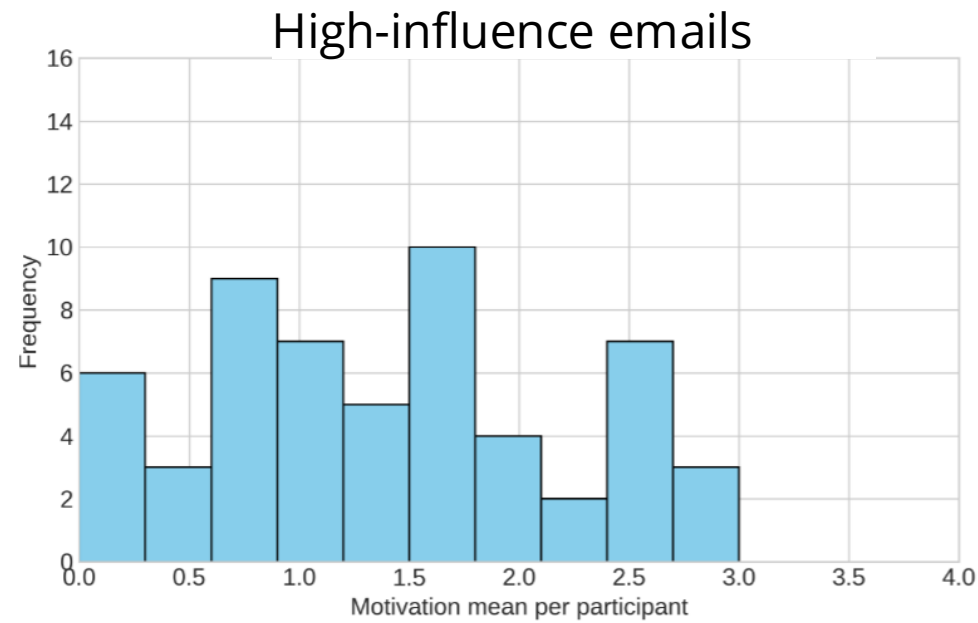


RESULTS & DISCUSSION

MOTIVATION

Hypothesis testing results:

- Testing showed no evidence of motivation frequency being different in both groups
- Testing showed evidence of different phishing classification rates in both groups



CLASSIFICATION ELEMENTS

ChatGPT-generated

- Emojis & Dashes
- Images

Not-ChatGPT-generated

- Familiarity
- Brand

Both

- Language
- Formatting
- Topic

GENERAL DISCUSSION

ANSWERING RESEARCH QUESTIONS

RESEARCH QUESTION

- Can LLMs generate **convincing** phishing HTML emails that accurately mimic authentic professional communications?

RQ1

- To what extent do participants misclassify emails? (ChatGPT/Not-ChatGPT)

RQ2

- What factors (e.g., professional formatting, images) contribute to the classification?

RQ3

- Is there a relation between the classification of ChatGPT/Not-ChatGPT with the classification of Phishing/Not-Phishing?

LIMITATIONS

Lack of
ecological
validity

Limited
device
diversity

No control
questions

Assumptions
about
ChatGPT

Performance
of ChatGPT

Single open
question per
email

CONCLUSION

Conclusion

LLMs are able to assist the automation of convincing phishing emails.

Lowering the skills necessary for creating high-quality phishing emails in scale.